MILITARY STANDARD

MIL-STD-1320-108
(NAVY)

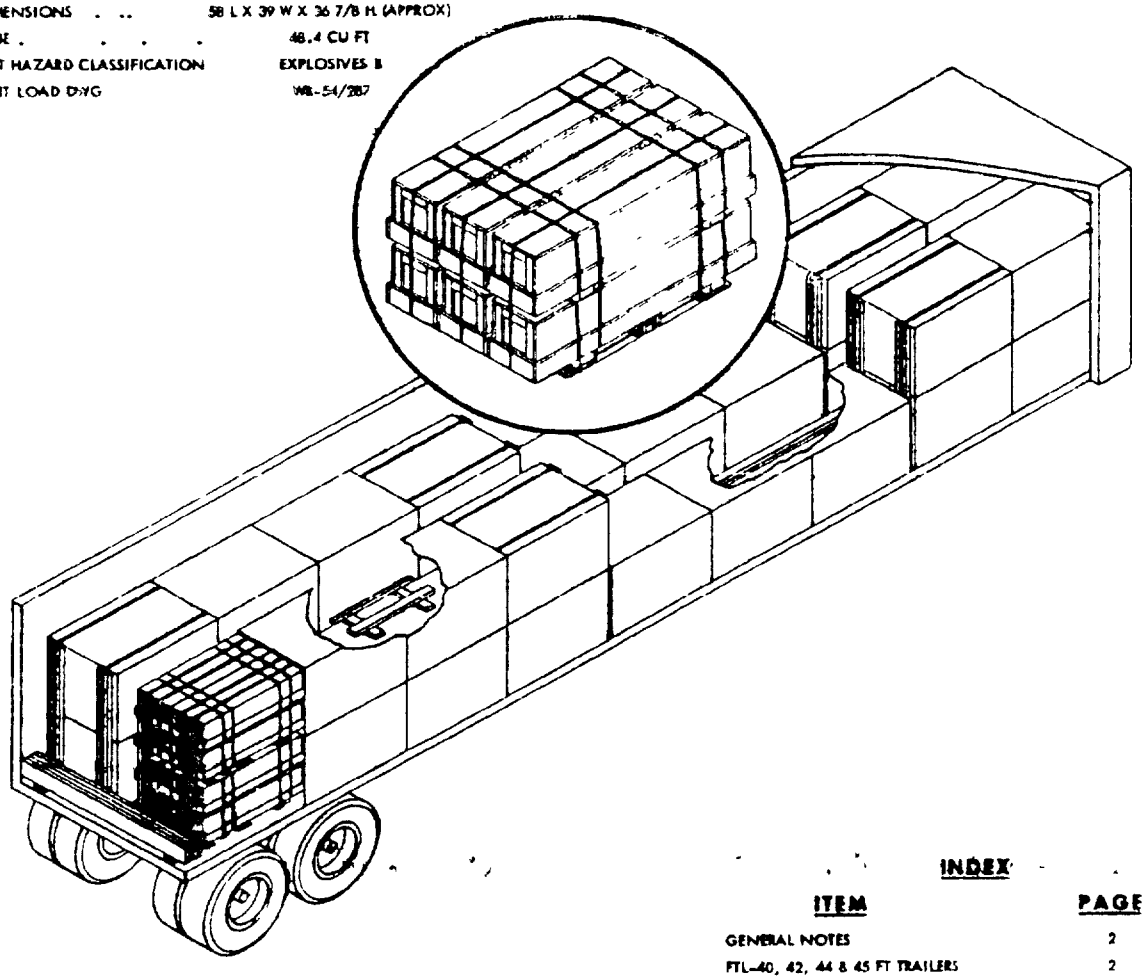
16 JUNE 1976

TRUCKLOADING

ROCKET MOTOR MK 52 AND MODS (SPARROW) OR, ROCKET MOTOR MK 53 MOD 1 OR MK 78 MOD 0 (SHRIKE) FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

WEIGHT (APPROX)	SEE TABLE I
DIMENSIONS . ..	58 L X 39 W X 36 7/8 H (APPROX)
CUBE	48.4 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B
UNIT LOAD DWG	MIL-54/287



INDEX

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GENERAL NOTES	2
FTL-40, 42, 44 & 45 FT TRAILERS	2
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DETAILS	6 & 7
UNITIZING DETAIL	8

NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-226 (NAVY)

FSC 8140

AUTHORIZED FOR HIGHWAY MOVEMENT ONLY

TECHNICAL DIRECTION AGENT (NDA) — DATE 4/6/76

AIRSYSCOM BY DIRECTION — DATE 5/3/76

ORIGINATOR — DATE 4/6/76

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF MK 52 MK 53 AND MK 78 ROCKET MOTORS WHEN THEY ARE PALLETIZED IN ACCORDANCE WITH WR-54/287

2 THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 & 45 FT VANS THE TRAILERS MAY HAVE WOOD FLOORS, METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS ALL TRAILERS SHALL HAVE THE TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER)

3 SWAY BRACE, DETAIL J, IS ASSEMBLED IN PLACE PIECE 6 IS INBOARD OF THE CORNER PALLET POSTS WITH PIECE 7 AGAINST THE PALLET POSTS, NAIL TO PIECE 6 WITH THREE 10d NAILS PER JOINT

4 TO PREVENT LONGITUDINAL MOVEMENT OF THE UNIT LOADS IN THE SECOND LAYER, THE TOP UNIT LOAD SHALL BE STRAPPED TO THE BOTTOM UNIT LOAD WHERE THE PATTERN CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH TWO HIGH UNIT LOADS AT THE REAR OF THE TRAILER SHALL ALSO BE STRAPPED TOGETHER SEE THE UNITIZING DETAIL, PAGE 8 FOR THE CORRECT PROCEDURE

5 THE TYPE OF BOLCKING AND BRACING TO USE AT THE REAR OF THE LOAD IS DEPENDENT UPON THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED DETERMINE THIS DISTANCE AND INSTALL BLOCKING AS INDICATED BELOW

DISTANCE LESS THAN 12 INCHES

INSTALL REAR BLOCKING DETAIL F PAGE 6
AS SHOWN ON PAGE 3

WARNING

DO NOT USE TRAILERS WITH ROLL UP REAR DOORS ASSEMBLY SHALL FILL THE VOID BETWEEN THE CLOSED DOORS AND LADING

DISTANCE 12 TO 36 INCHES

INSTALL REAR BLOCKING DETAIL D PAGE 6
AS SHOWN ON PAGE 5

WARNING

DO NOT USE TRAILERS WITH ROLL UP REAR DOORS ASSEMBLY SHALL FILL THE VOID BETWEEN THE DOORS AND LADING

DISTANCE GREATER THAN 36 INCHES

INSTALL KNEE BRACE DETAIL K PAGE 7
AS SHOWN ON PAGE 7

WARNING

DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

6 THE LENGTH OF THE STRUT, PIECE 9, SHALL BE SUFFICIENT TO CAUSE THE REAR BLOCKING ASSEMBLY, DETAIL D TO BEAR AGAINST THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED

7 THE QUANTITY AND THICKNESS OF THE MATERIAL USED IN THE REAR BLOCKING ASSEMBLY, DETAIL F , MAY BE VARIED THE BLOCKING SHALL FILL THE VOID BETWEEN THE LADING AND THE REAR DOORS WHEN THE DOORS ARE CLOSED

8 WHEN A UNIT LOAD IN THE SECOND LAYER DOES NOT HAVE A UNIT LOAD OPPOSITE IT THE UNITIZED LOAD SHALL BE STRAPPED TO PREVENT LATERAL MOTION USING TWO 1 1/4 X .035 STRAPS AND SEAL THE STRAPS ENCIRCLE THE STACKED LOAD PASSING UNDER THE PALLET DECK OF THE BOTTOM UNIT LOAD AND OVER THE TOP OF THE TOP UNIT LOAD SEE PIECES 15 & 16, PAGE 5

9 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

10 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVE B" PLACARD TO BOTH SIDES FRONT AND BACK OF TRAILER

11 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER, MM-L-751, NAILS, FF-N-105, TYPE II STYLE 98, COMMON WEIGHT, STRAPPING QQ-S-781, TYPE I CLASS A

12 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY)

FTL - 40, 42, 44, & 45 FT TRAILERS (VAN)

1 A FULL TRUCKLOAD CONSISTS OF

MK 52 & MK 53 ROCKET MOTORS . . . . . 30 UNIT LOADS

MK 78 ROCKET MOTORS . . . . . 28 UNIT LOADS

WHEN LOADING 40, 42 & 44 FT TRAILERS, USE THE CORRECT LOAD PATTERN AS SHOWN IN TABLE II, PAGE 4 WHEN LOADING 45 FT TRAILERS, USE THE CORRECT LOAD PATTERN AS SHOWN IN TABLE III, PAGE 4.

NOTE

A) 40 & 45 FT VANS DO NOT REQUIRE A FRONT BULKHEAD IF THE VAN HAS A SQUARE CORNER FRONT WALL, POSITION UNIT LOADS AGAINST WALL AS SHOWN ON PAGE 3. IF THE FRONT WALL HAS ROUNDED CORNERS, POSITION AGAINST THE FRONT WALL AS SHOWN ON PAGE 7 BLOCK THE REAR OF THE LOAD USING DETAIL F AS SHOWN ON PAGE 3

B) 42 & 44 FT VANS REQUIRE THAT THE FRONT BULKHEAD, DETAIL G, PAGE 6, BE USED INSTALL USING THE PRINCIPLES SHOWN ON PAGE 5. BLOCK THE REAR OF THE LOAD USING DETAIL D AS SHOWN ON PAGE 5.

TABLE I

FULL TRUCKLOAD DATA

ROCKET MOTOR			UNIT LOAD	NUMBER	FTL
DODC/ NALC NO	MK	ON PALLET MK & MOD	WT (APPROX) LBS	UNIT LOADS FTL	WT (APPROX) LBS
V357	52	12 - 0	1412	30	42,360
V357	52	12 - 1	1396	30	41,880
V334	53	12 - 0	1434	30	42,720
V334	53	12 - 1	1386	30	41,580
V332	78	12 - 0	1514	28	42,392
V388	78	12 - 1	1476	28	41,328

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NOTE

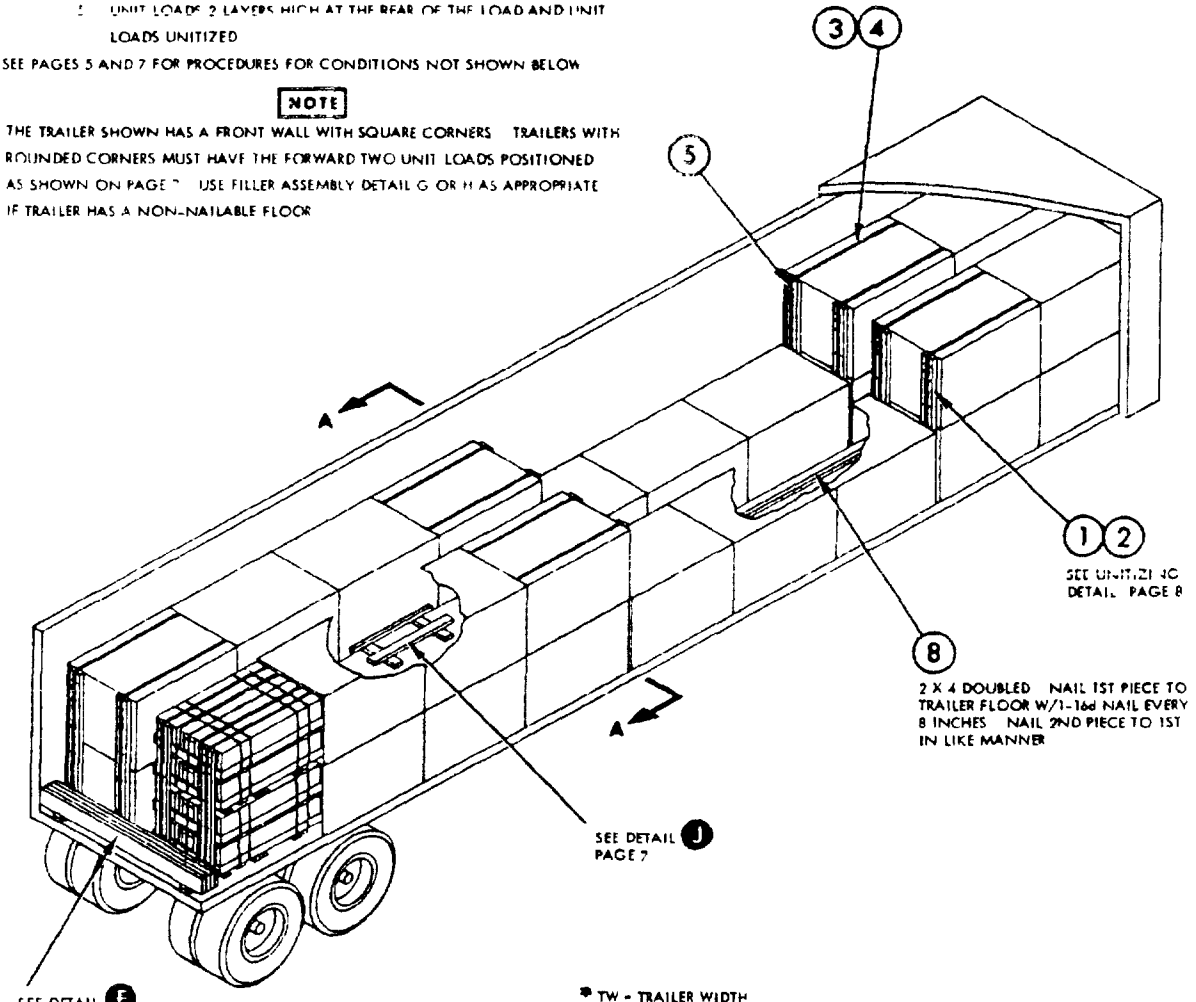
THIS TRUCKLOAD SHOWS THE PROCEDURES FOR THE FOLLOWING CONDITIONS

1 TRAILER WITH A NAILABLE FLOOR (WOOD OR METAL AND WOOD
2 SQUARE NOSE TRAILER WITH NO BULKHEAD
3 SPACE AT REAR OF LOAD LESS THAN 12 INCHES
4 PARTIAL SECOND LAYER WITH LAYERS CHANGING FROM 2 TO 1 LAYER HIGH AND TWO HIGH UNIT LOADS UNITIZED
5 UNIT LOADS 2 LAYERS HIGH AT THE REAR OF THE LOAD AND UNIT LOADS UNITIZED

SEE PAGES 5 AND 7 FOR PROCEDURES FOR CONDITIONS NOT SHOWN BELOW

NOTE

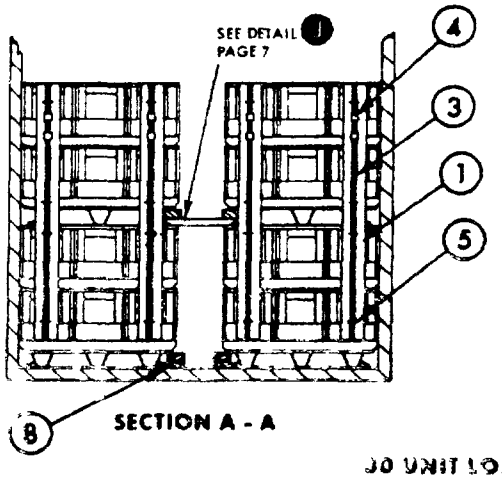
THE TRAILER SHOWN HAS A FRONT WALL WITH SQUARE CORNERS TRAILERS WITH ROUNDED CORNERS MUST HAVE THE FORWARD TWO UNIT LOADS POSITIONED AS SHOWN ON PAGE 7 USE FILLER ASSEMBLY DETAIL G OR H AS APPROPRIATE IF TRAILER HAS A NON-NAILABLE FLOOR



* TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
10	SUPPORT	2 X 4 TO SUIT	4	9 / 10	3/JOINT	10d / 10d
9	CROSSMEMBER	2 X 8 X TW -1 *	AS REQ'D	LAMINATE	AS REQ'D	AS REQ'D
8	SLEEPER	2 X 4 X 46	36	SEE FIELD NOTE		
7	STRINGER	2 X 4 X 48	12	6	3/JOINT	10d
6	SUPPORT	2 X 4 X 30	12	SEE 7	-	-
5	STAPLE, STRAP	FOR 1 1/4 STRAP	48	-	-	-
4	SEAL	FOR 1 1/4 STRAP	24	-	-	-
3	STRAP	1 1/4 X .035 X 23 FT	12	-	-	-
2	SPACER	1 X 2 X 5 1/2	12	1	3	6d
1	STIFFENER	2 X 6 X 68 3/4	12	-	-	-

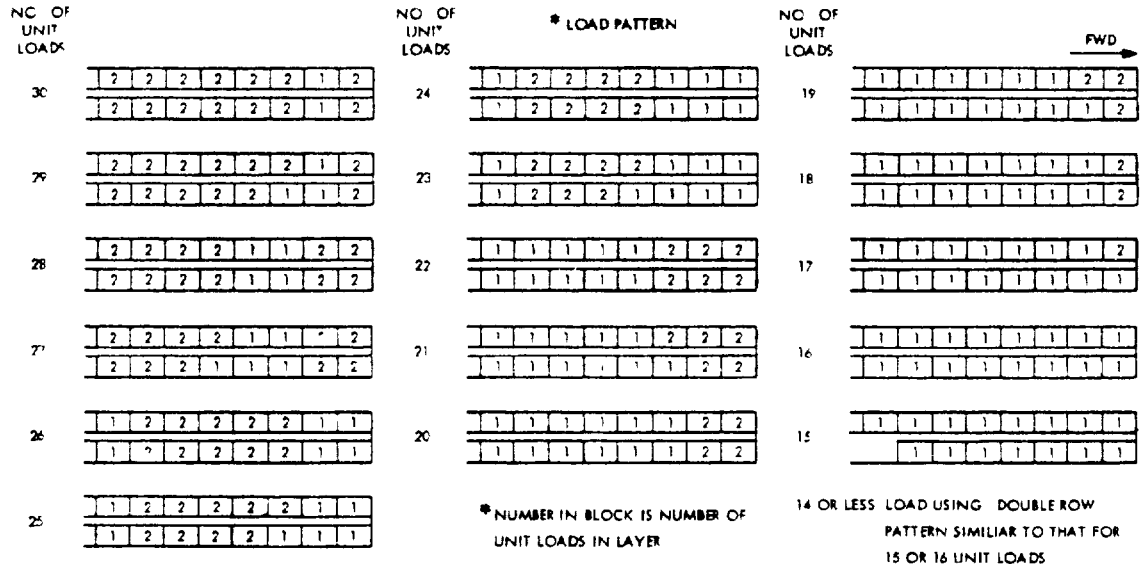
LIST OF MATERIALS & NAILING DATA



SECTION A - A

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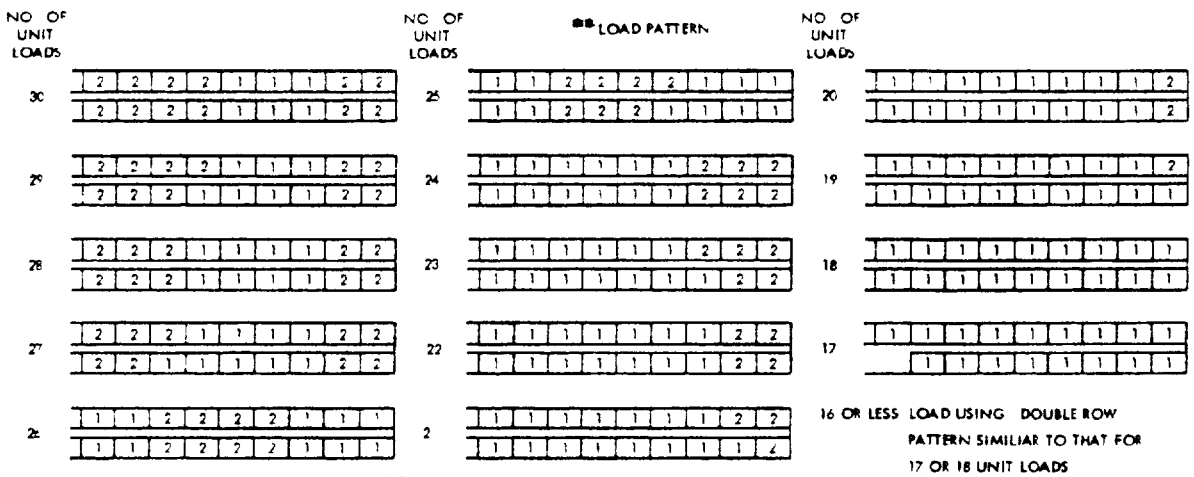
TABLE II
40, 42 & 44 FT TRAILERS



*NUMBER IN BLOCK IS NUMBER OF UNIT LOADS IN LAYER

14 OR LESS LOAD USING DOUBLE ROW PATTERN SIMILIAR TO THAT FOR 15 OR 16 UNIT LOADS

TABLE III
45 FT TRAILERS



16 OR LESS LOAD USING DOUBLE ROW PATTERN SIMILIAR TO THAT FOR 17 OR 18 UNIT LOADS

**NUMBER IN BLOCK IS NUMBER OF UNIT LOADS IN LAYER

LTL - 40, 42, 44 & 45 FT TRAILERS (VAN)

1 WHEN REQUIRED TO SHIP LESS-THAN TRUCKLOAD USE TABLE II FOR 40 42 & 44 FT TRAILERS AND TABLE III FOR 45 FT TRAILERS BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 3, 5 AND 7

NOTE

A) 40 & 45 FT VANS DO NOT REQUIRE A FRONT BULKHEAD IF THE VAN HAS A SQUARE CORNER FRONT WALL, POSITION UNIT LOADS AGAINST WALL AS SHOWN ON PAGE 3 IF FRONT WALL HAS ROUNDED CORNERS, POSITION AGAINST THE FRONT WALL AS SHOWN ON PAGE 7

B) 42 & 44 FT VANS REQUIRE THAT THE APPROPRIATE FRONT BULKHEAD BE USED WHEN LOADING 17 OR MORE UNIT LOADS WHEN LOADING 16 OR LESS UNIT LOADS, A FRONT BULKHEAD IS NOT REQUIRED

WARNING

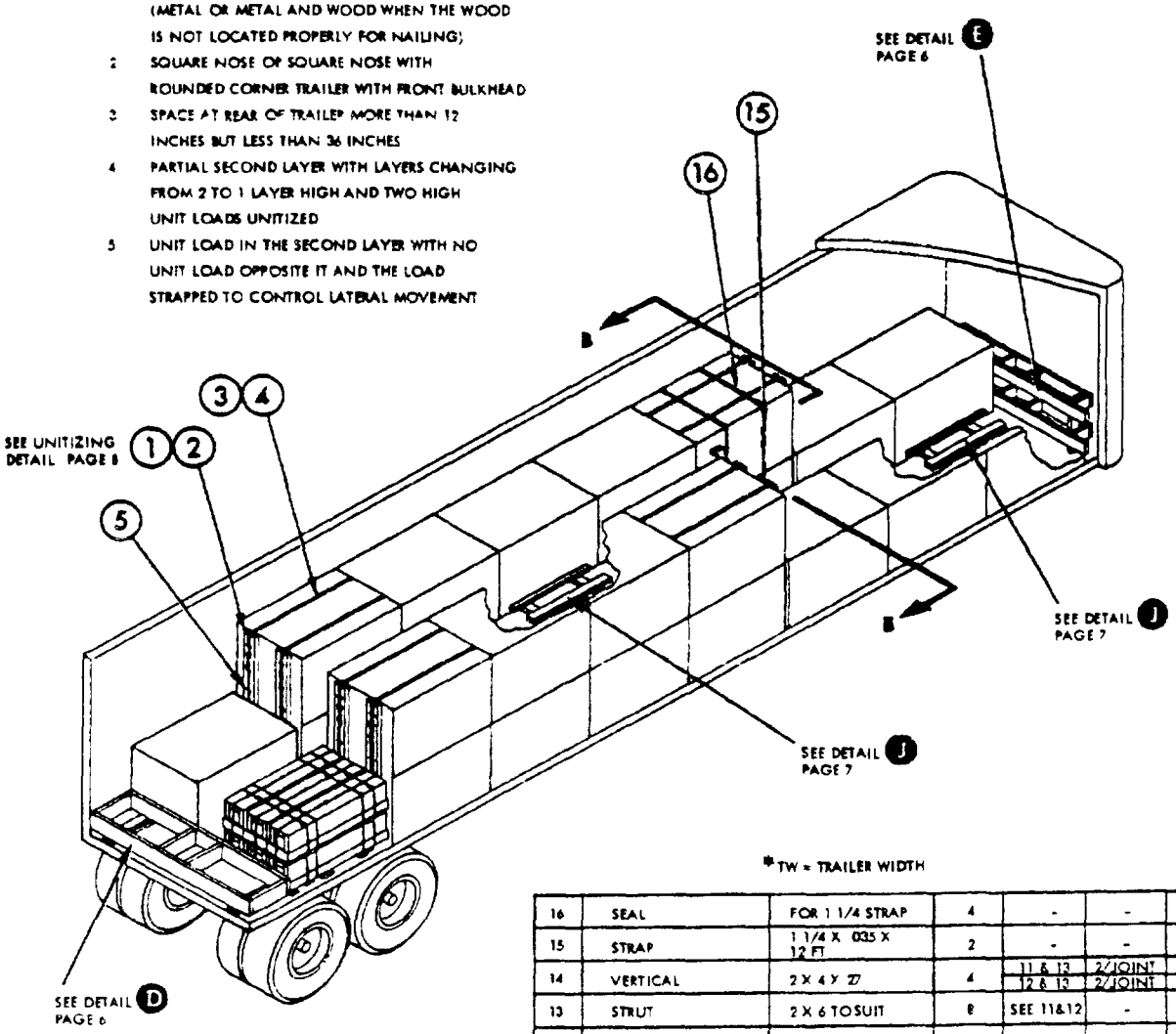
DO NOT NAIL INTO METAL NON-NAILABLE FLOOR IF BLOCKING REQUIRES NAILING INTO THE TRAILER FLOOR, METAL OR NON-NAILABLE FLOOR TRAILERS CAN NOT BE USED

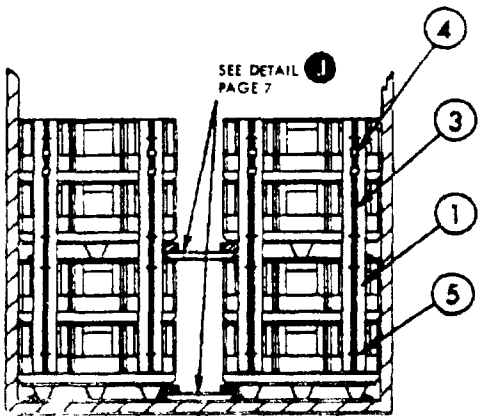
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NOTE

THIS TRUCKLOAD SHOWS THE PROCEDURES FOR THE FOLLOWING CONDITIONS

1. TRAILER WITH A NON-NAILABLE FLOOR (METAL OR METAL AND WOOD WHEN THE WOOD IS NOT LOCATED PROPERLY FOR NAILING)
2. SQUARE NOSE OR SQUARE NOSE WITH ROUNDED CORNER TRAILER WITH FRONT BULKHEAD
3. SPACE AT REAR OF TRAILER MORE THAN 12 INCHES BUT LESS THAN 36 INCHES
4. PARTIAL SECOND LAYER WITH LAYERS CHANGING FROM 2 TO 1 LAYER HIGH AND TWO HIGH UNIT LOADS UNITIZED
5. UNIT LOAD IN THE SECOND LAYER WITH NO UNIT LOAD OPPOSITE IT AND THE LOAD STRAPPED TO CONTROL LATERAL MOVEMENT



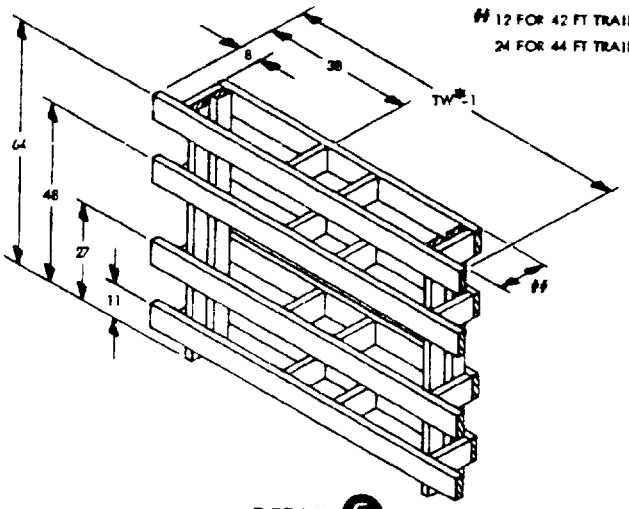


SECTION B-B

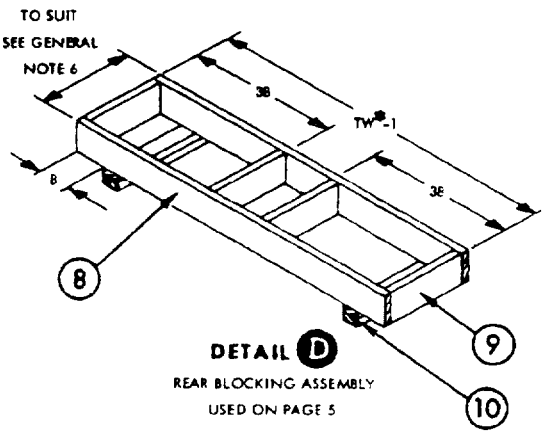
*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS & NAILING DATA						
1	STIFFENER	2 X 6 X 68 3/4	8	-	-	-
2	SPACER	1 X 2 X 5 1/2	8	-	-	-
3	STRAP	1 1/4 X .035 X 23 FT	8	-	-	-
4	SEAL	FOR 1 1/4 STRAP	16	-	-	-
5	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
6	SUPPORT	2 X 4 X 30	24	SEE 7	-	-
7	STRINGER	2 X 4 X 48	24	6	3/JOINT	10d
8	CROSSMEMBER	2 X 8 X TW -1*	2	SEE 9	-	-
9	STRUT	2 X 8 TO SUIT	4	8	3/JOINT	16d
10	SUPPORT	2 X 4 X TO SUIT	4	8 / 10	3/JOINT / 3	10d / 10d
11	AFT HORIZONTAL	2 X 6 X TW -1*	2	13	3/JOINT	16d
12	FWD HORIZONTAL	2 X 6 TO SUIT	2	13	3/JOINT	16d
13	STRUT	2 X 6 TO SUIT	8	SEE 11&12	-	-
14	VERTICAL	2 X 4 X 27	4	11 & 13 / 12 & 13	2/JOINT / 2/JOINT	10d / 10d
15	STRAP	1 1/4 X .035 X 12 FT	2	-	-	-
16	SEAL	FOR 1 1/4 STRAP	4	-	-	-

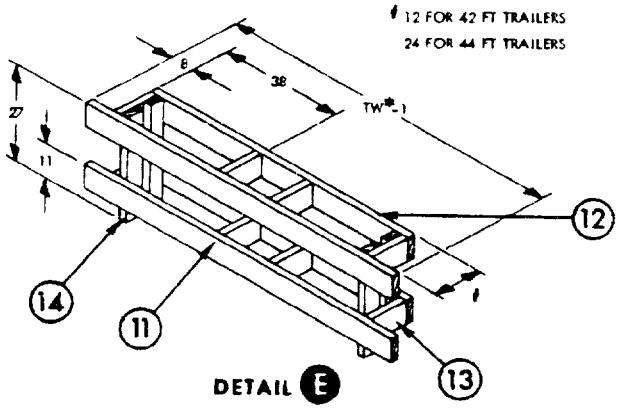
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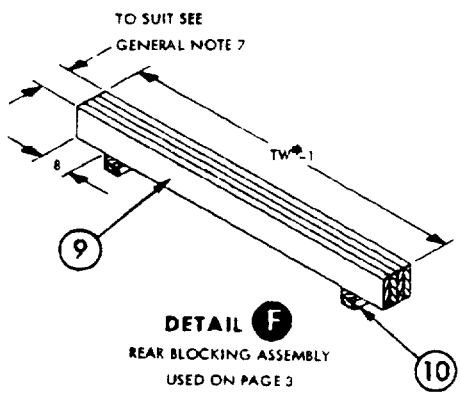
DETAIL C
FRONT BULKHEAD
(REQUIRED FOR 2 HIGH LOAD)



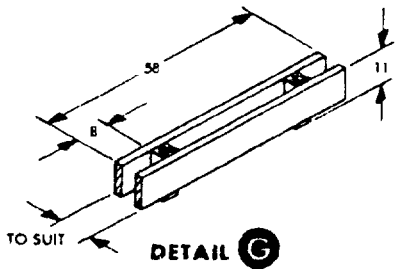
DETAIL D
REAR BLOCKING ASSEMBLY
USED ON PAGE 5



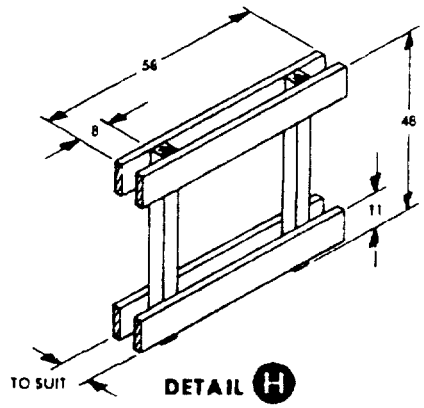
DETAIL E
FRONT BULKHEAD
USED ON PAGE 5



DETAIL F
REAR BLOCKING ASSEMBLY
USED ON PAGE 3



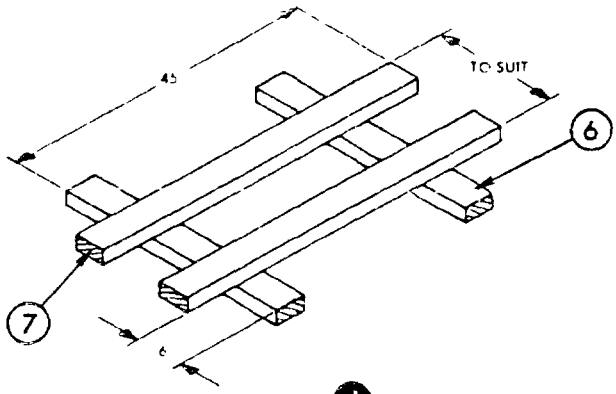
DETAIL G
FILLER ASSEMBLY
SEE NOTE ON ROUNDED
CORNERS, PAGE 3



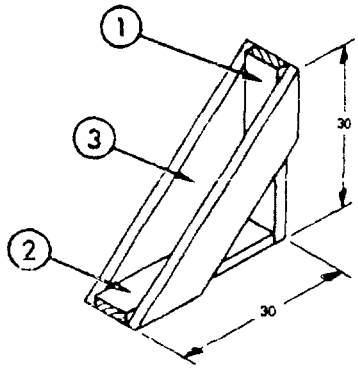
DETAIL H
FILLER ASSEMBLY
SEE NOTE ON ROUNDED
CORNERS PAGE 3

*TW - TRAILER WIDTH

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DETAIL J
SWAY BRACE
(USED ON PAGES 3 & 5



WITH BRACE IN POSITION NAIL PIECE 2 TO TRAILER FLOOR WITH EIGHT 16d NAILS PER BRACE

DETAIL K
KNEE BRACE

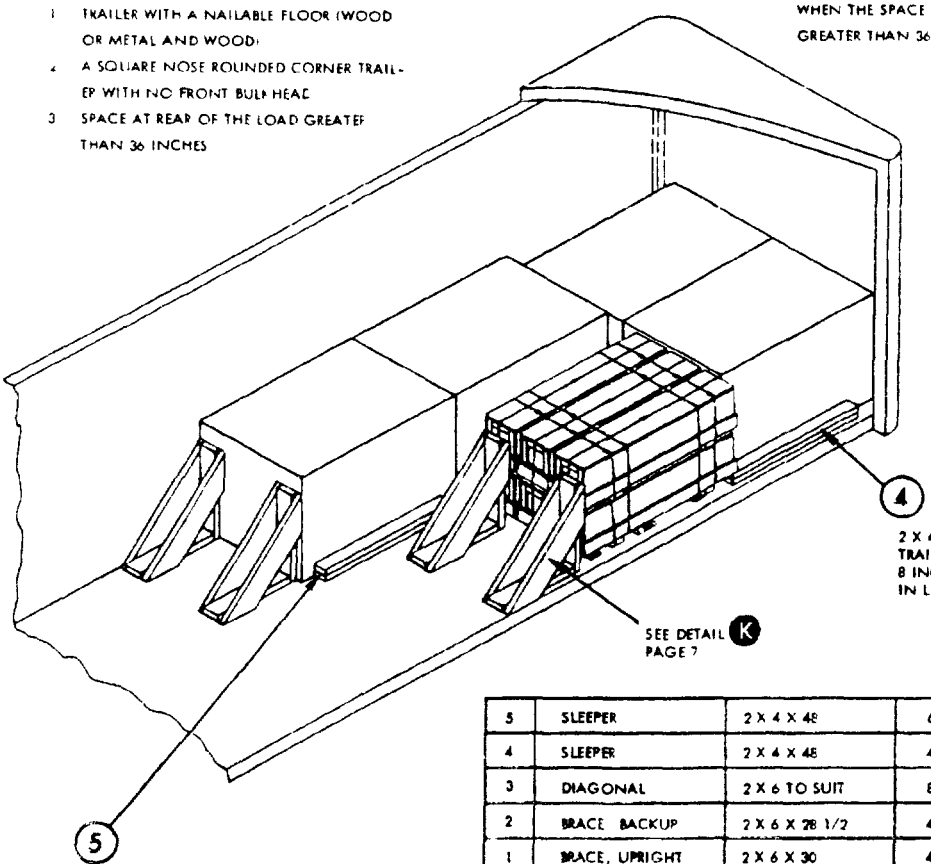
NOTE

THIS LTL SHOWS THE PROCEDURES FOR THE FOLLOWING CONDITIONS

1. TRAILER WITH A NAILABLE FLOOR (WOOD OR METAL AND WOOD)
2. A SQUARE NOSE ROUNDED CORNER TRAILER WITH NO FRONT BULKHEAD
3. SPACE AT REAR OF THE LOAD GREATER THAN 36 INCHES

WARNING

DO NOT USE TRAILERS WITH NON-NAILABLE FLOOR WHEN THE SPACE AT THE REAR OF THE LOAD IS GREATER THAN 36 INCHES



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

SEE DETAIL K PAGE 7

2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

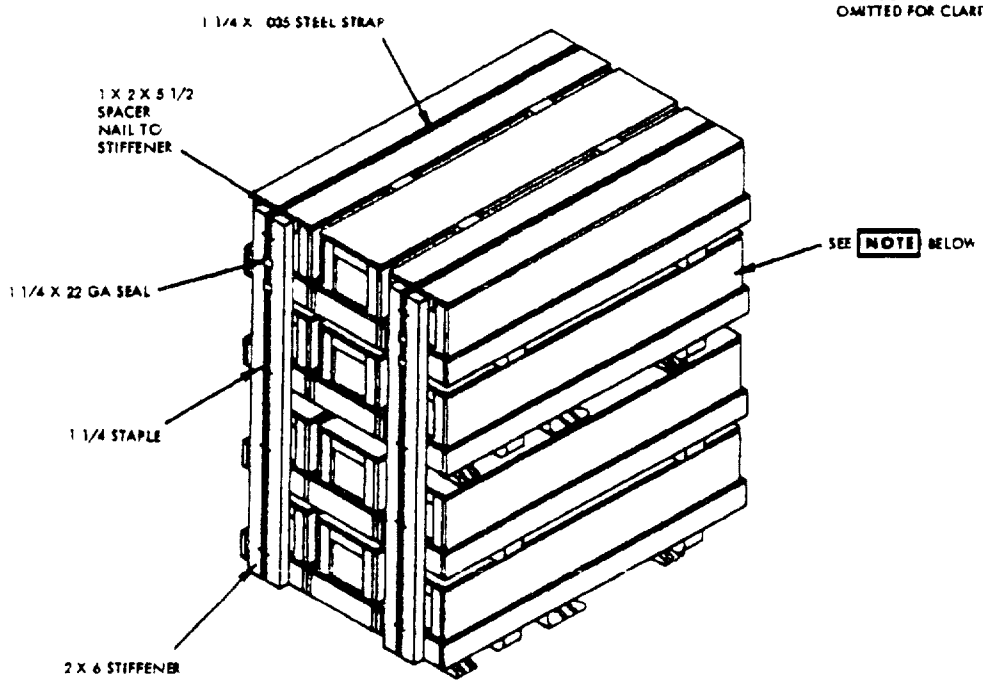
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
5	SLEEPER	2 X 4 X 48	6	SEE FIELD NOTE		
4	SLEEPER	2 X 4 X 48	4	SEE FIELD NOTE		
3	DIAGONAL	2 X 6 TO SUIT	8	1 & 2	4 PER JOINT	16d
2	BRACE BACKUP	2 X 6 X 28 1/2	4	SEE DETAIL K		
1	BRACE, UPRIGHT	2 X 6 X 30	4	2	3 PER JOINT	16d

LIST OF MATERIALS & NAILING DATA

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UTILIZING DETAIL

NOTE
UNIT LOAD STRAPS OMITTED FOR CLARITY



1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, THE STACKED UNIT LOADS SHALL BE UNITIZED AS SHOWN ABOVE
2. THE 2 X 6 STIFFENER SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD A 1 X 2 X 5 1/2 SPACER SHALL BE AT THE TOP INBOARD SIDE OF EACH STIFFENER
3. THE 1 1/4 INCH STEEL STRAPS, POSITIONED AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET THE STRAPS HOLD THE STIFFENERS IN PLACE
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES

NOTE
WHEN LOADING A TRAILER WITH A SINGLE STACK OF TWO HIGH UNIT LOADS, THE STIFFENER SHALL BE AT BOTH ENDS OF THE UNITIZED LOADS

REVIEW ACTIVITY:
NAVY - OS, AS

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NAVY - OS
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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1320-108 (Navy)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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COMMANDING OFFICER
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NAVAL WEAPONS HANDLING LABORATORY (803)
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